Specification for deployment of self-developed chassis charging piles

- The charging pile needs to be placed on the wall, and there should be no other objects at each half meter position from the center of the charging pile. (see the picture)
- The position of charging piles shall be affixed with the indicating label stickers of charging piles as required.

Specification for deployment of self-developed chassis charging piles

- Charging piles need to be placed on flat ground, and can not be placed on slopes or carpet environment.
- The height of the charging sheet of the charging pile should be at the same height as the charging contact on the machine. If there is any error, the bottom of the charging pile should be adjusted to appropriately increase or reduce the filler.

Specification for deployment of self-developed chassis charging piles

The position of charging piles should be fixed relative to the scene to avoid movement due to personnel moving or other reasons in the later stage. Can be to paste marks on the ground or fixed by double-sided tape. (see the picture)

Specification for deployment of self-developed chassis charging piles

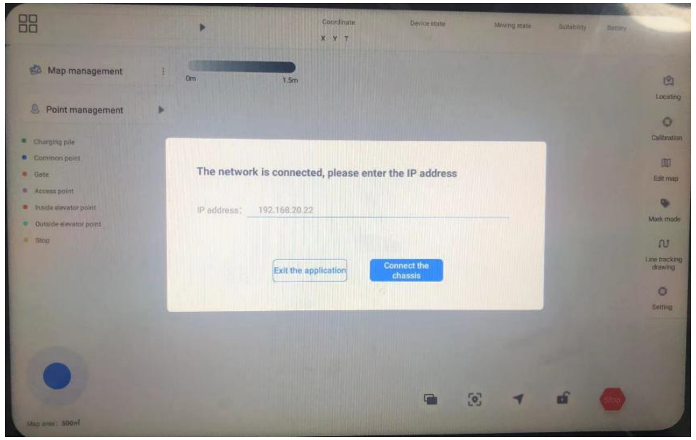
When multiple machines are used at the same time in the same scene, the distance between charging piles shall not be less than 1.5m. (see the picture)

General purpose (self - developed) chassis drawing tool

Two ways of connection of general (self-developed) chassis

The first (directly connected to the upper body) : install self-developed and deployed apps on Android in the machine body.

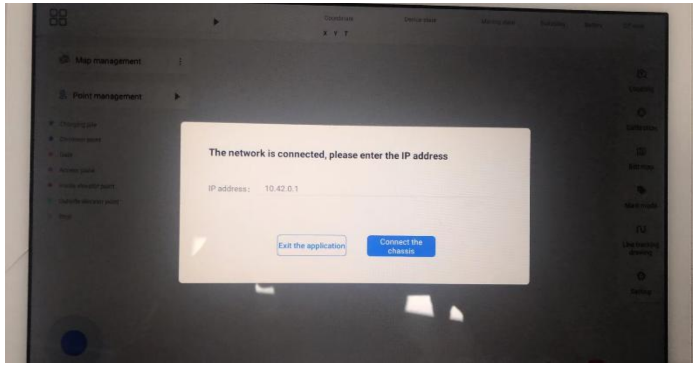
After opening the software, connect the chassis directly through 192.168.20.22.



Second (remote connection) : Use external Android devices (phone, tablet, win11 Android) Android.

Self-developed and deployed APP.

Connect the device to the WiFi of the chassis first, open the software, and remotely connect to the chassis through 10.42.0.1.



Remote WiFi description:

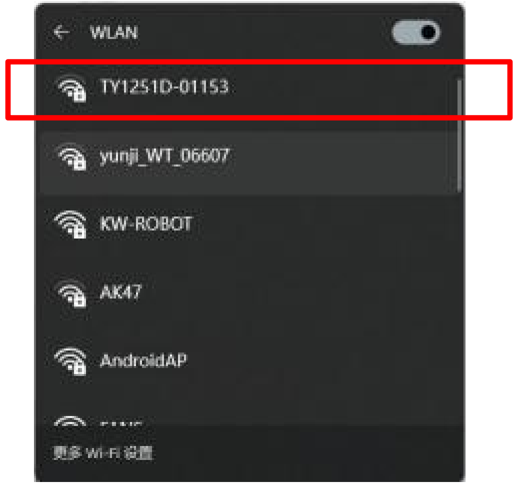
WiFi Name: TY1251D-xxxxx (last five digits of machine chassis code)

Initial password: 123456789

Remote connection IP address: 10.42.0.1

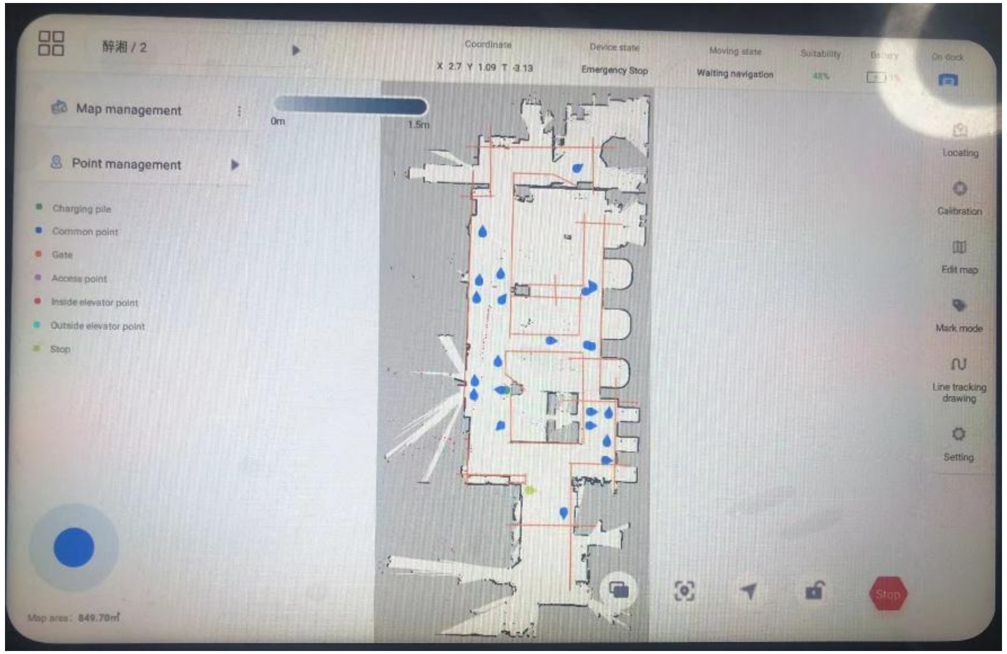
The default IP address for direct upper-body connection is 192.168.20.22

TY1251D-01153



General purpose (self - developed) chassis drawing tool

■ **Self-developed deployment APP page description**

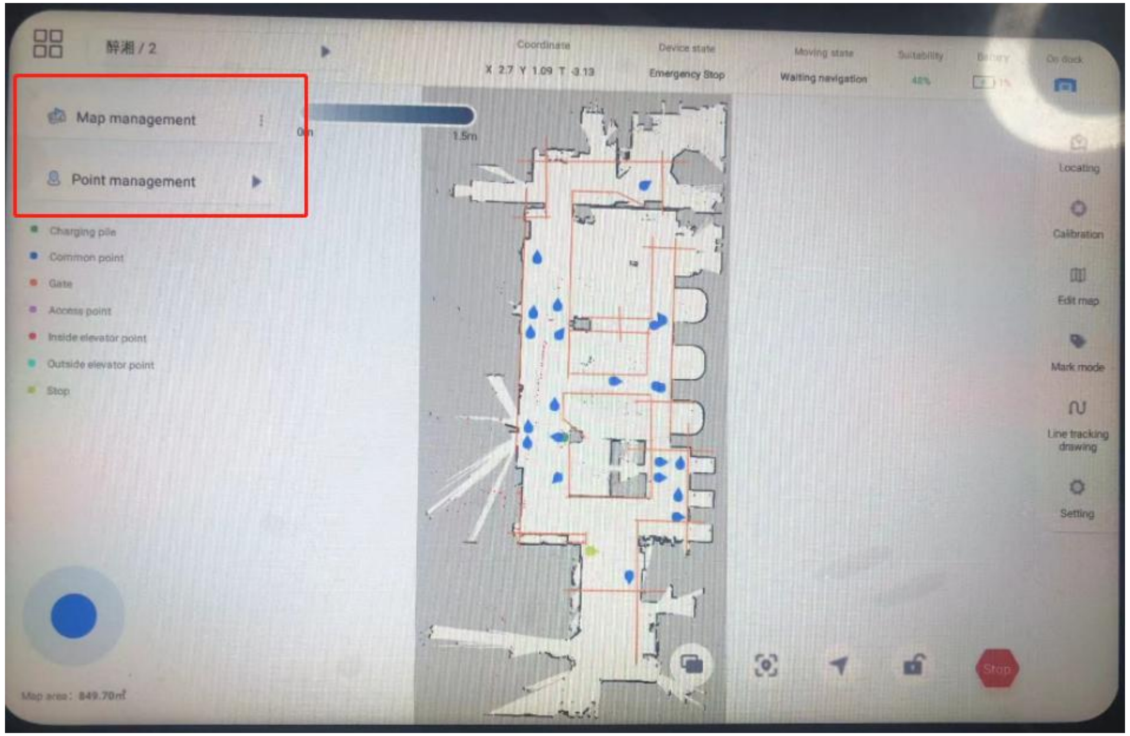


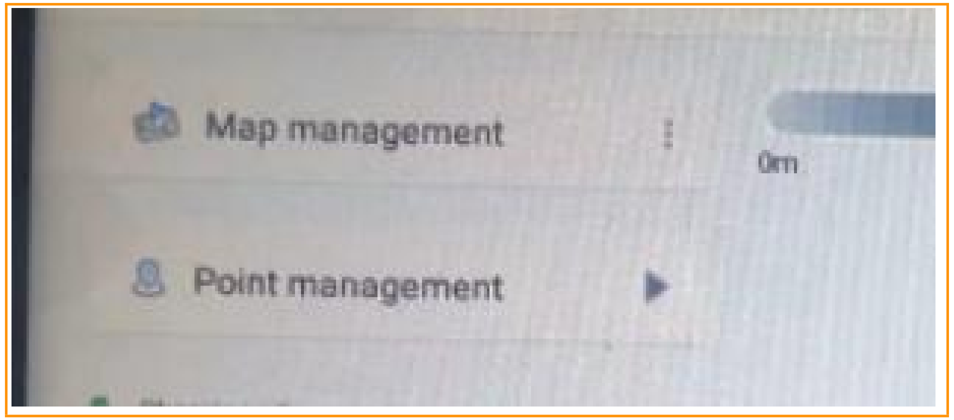
The machine status information, new map and map selection are displayed at the top Robot state information includes current position, equipment state, motion state, matching degree and electric quantity.

Click on the left button， There are new maps and map scan respectively. The algorithm version 4.6.1 and above supports map scan function. The New map function is displayed. Click the map name to enter the list to switch the map graphic displayed in the main map. Click the map to continue scanning, you can continue sweeping the map.

General purpose (self - developed) chassis drawing tool

- **Self-developed deployment APP page description**





The left side is map management and point management

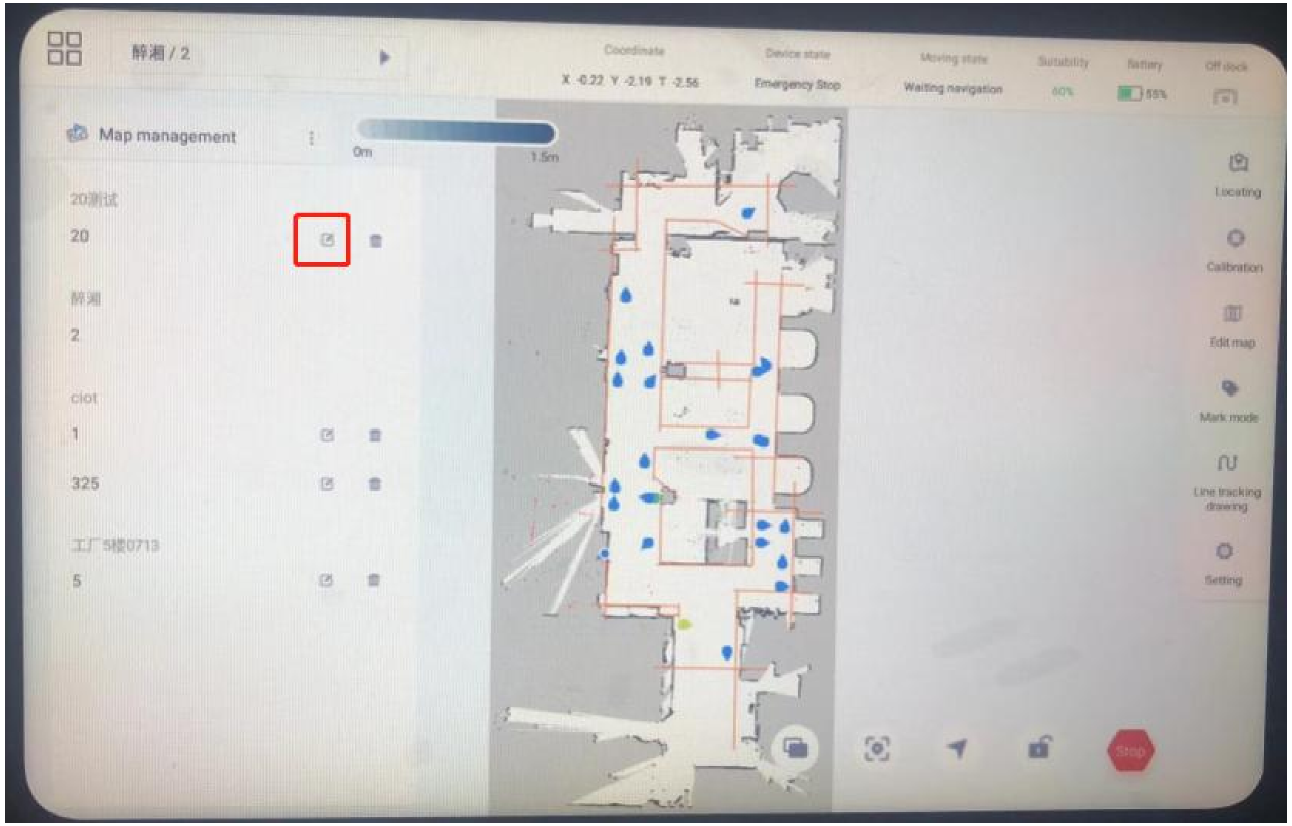
Click [Map Management] to expand the map list.

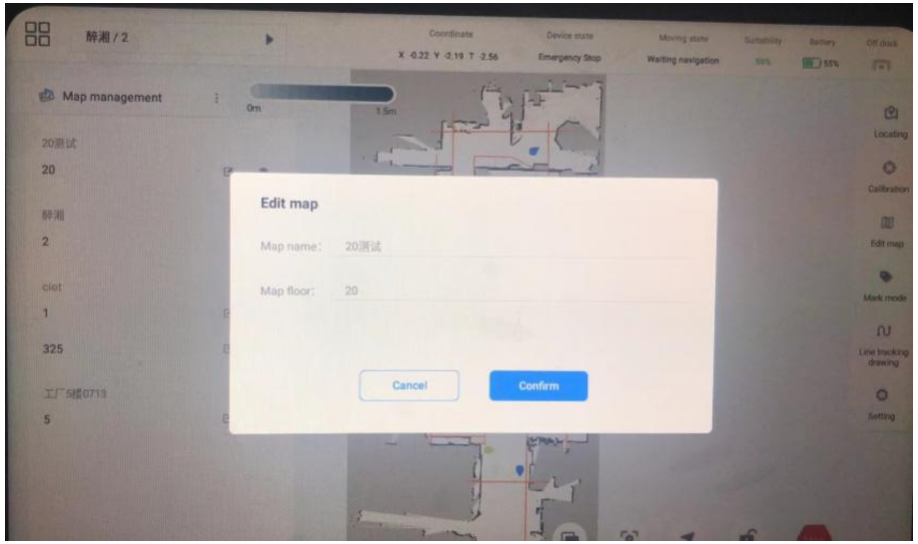
Click [Point Management] to expand the point list.

General purpose (self - developed) chassis drawing tool

■ Map management module

Click "Map Management" to expand the map list, and you can edit and delete the map.





Click the button，

The Modify Map page is displayed.

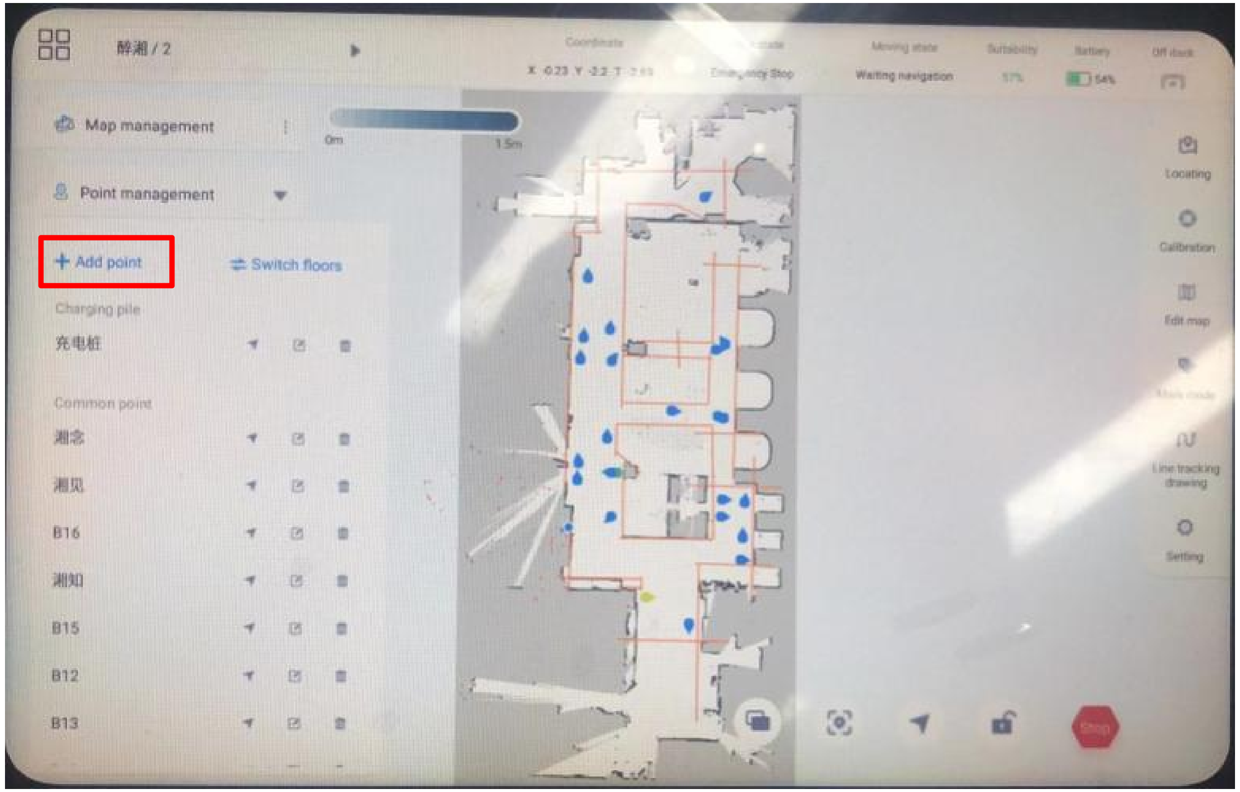
You can modify [map name] and [Map floor] information.

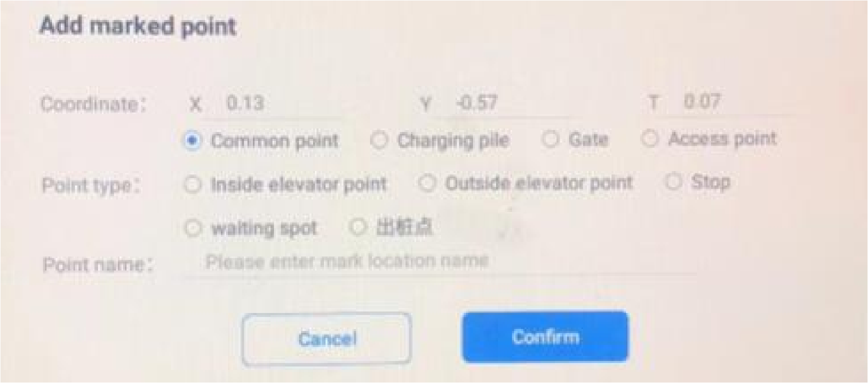
General purpose (self - developed) chassis drawing tool

■ Point management module

Click "point Management" to expand the point list and support the functions of adding point, point navigation, point editing, point deletion and cross-floor navigation.

Click "Add point position" and enter the information of added mark point position, as shown in the figure. Click "OK" after the input is completed.





Point specification:

1. At present, only one charging pile can be set in one diagram to prevent conflicts

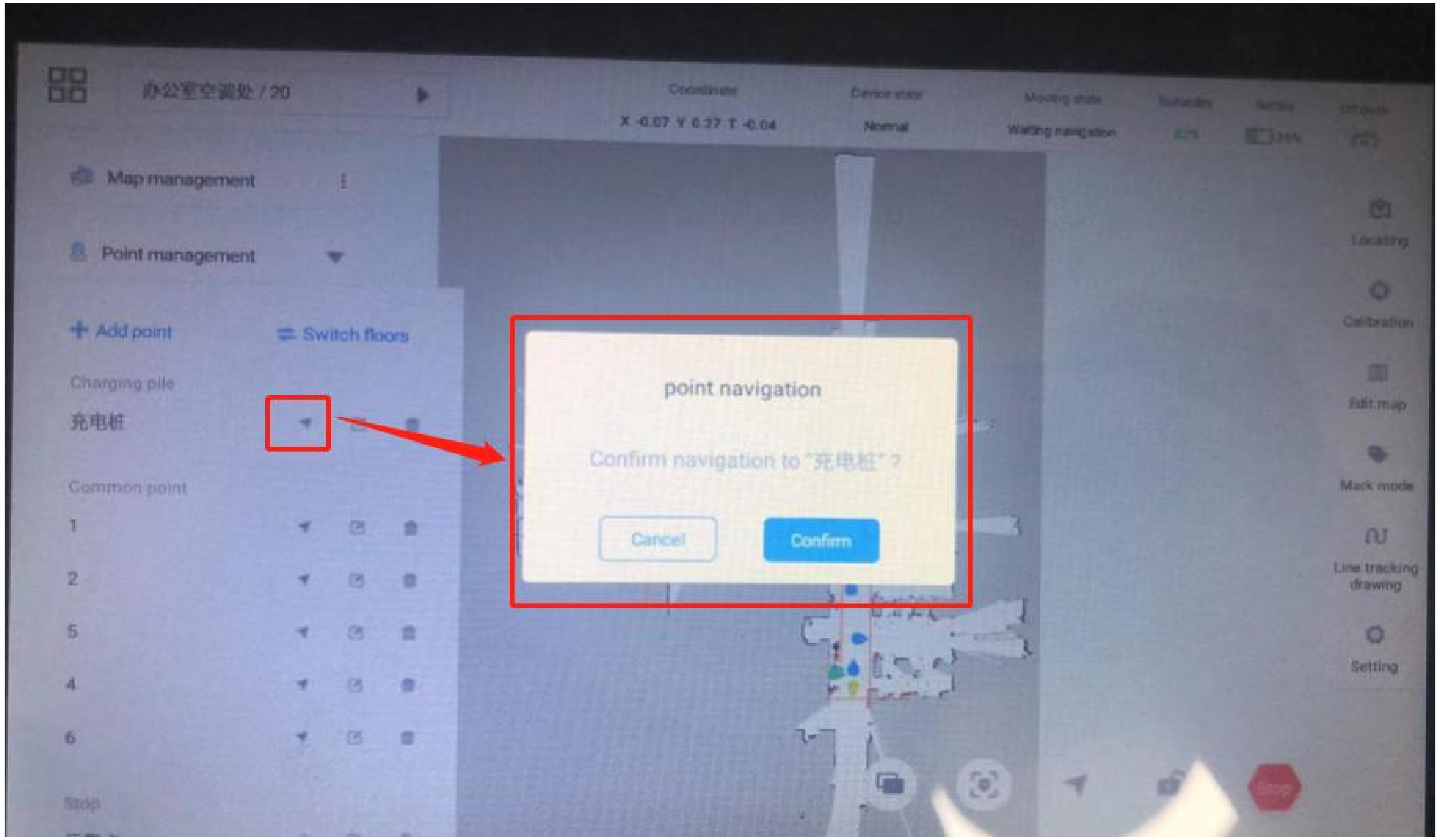
2. The gate access control in the point must be matched with the corresponding hardware module (default is not available).

3. The inside and outside of the elevator at the point need to match the corresponding hardware module (no default)

General purpose (self - developed) chassis drawing tool

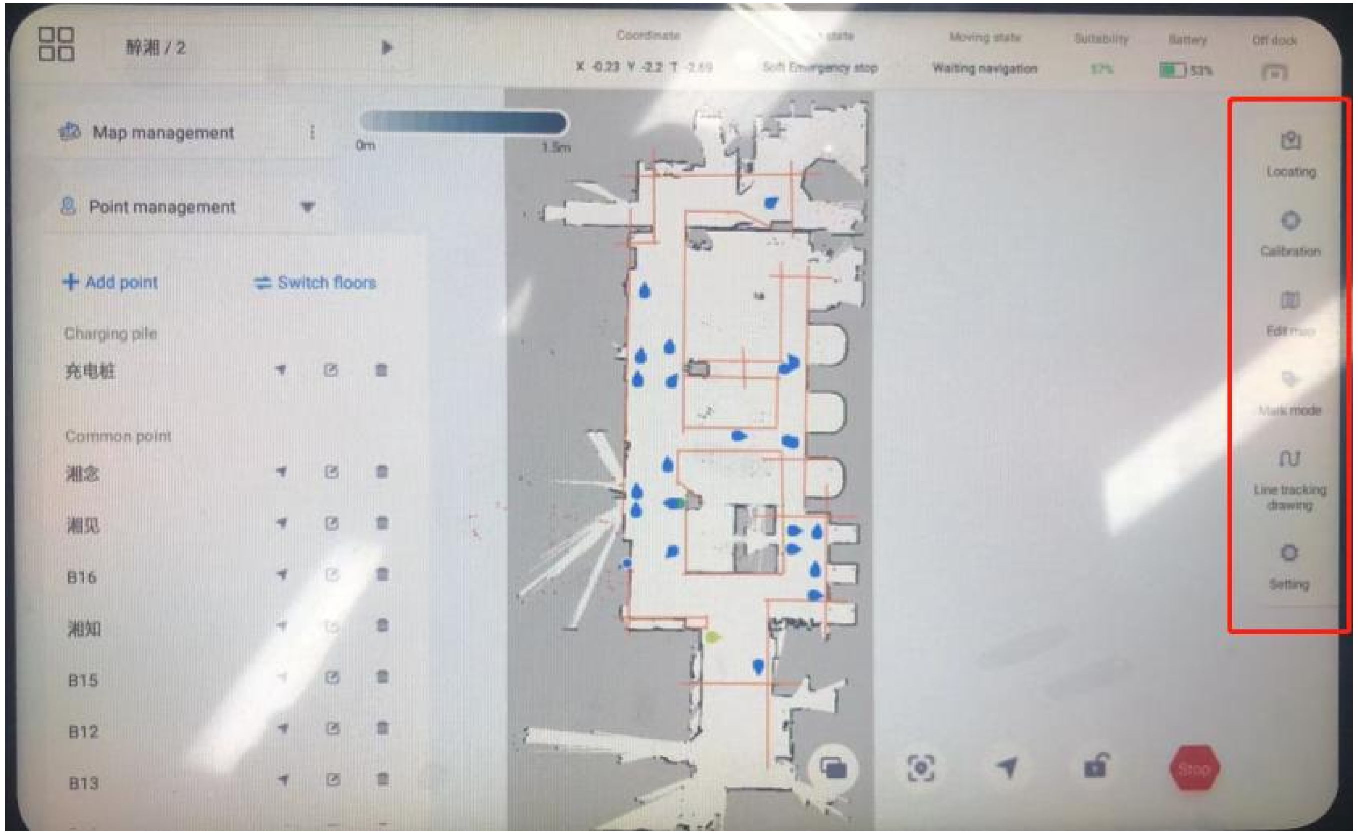
■ Point navigation

Select the corresponding point in the point list and click the button to navigate to the point.



General purpose (self - developed) chassis drawing tool

- **Self-developed deployment APP page description**



On the right are map related functions

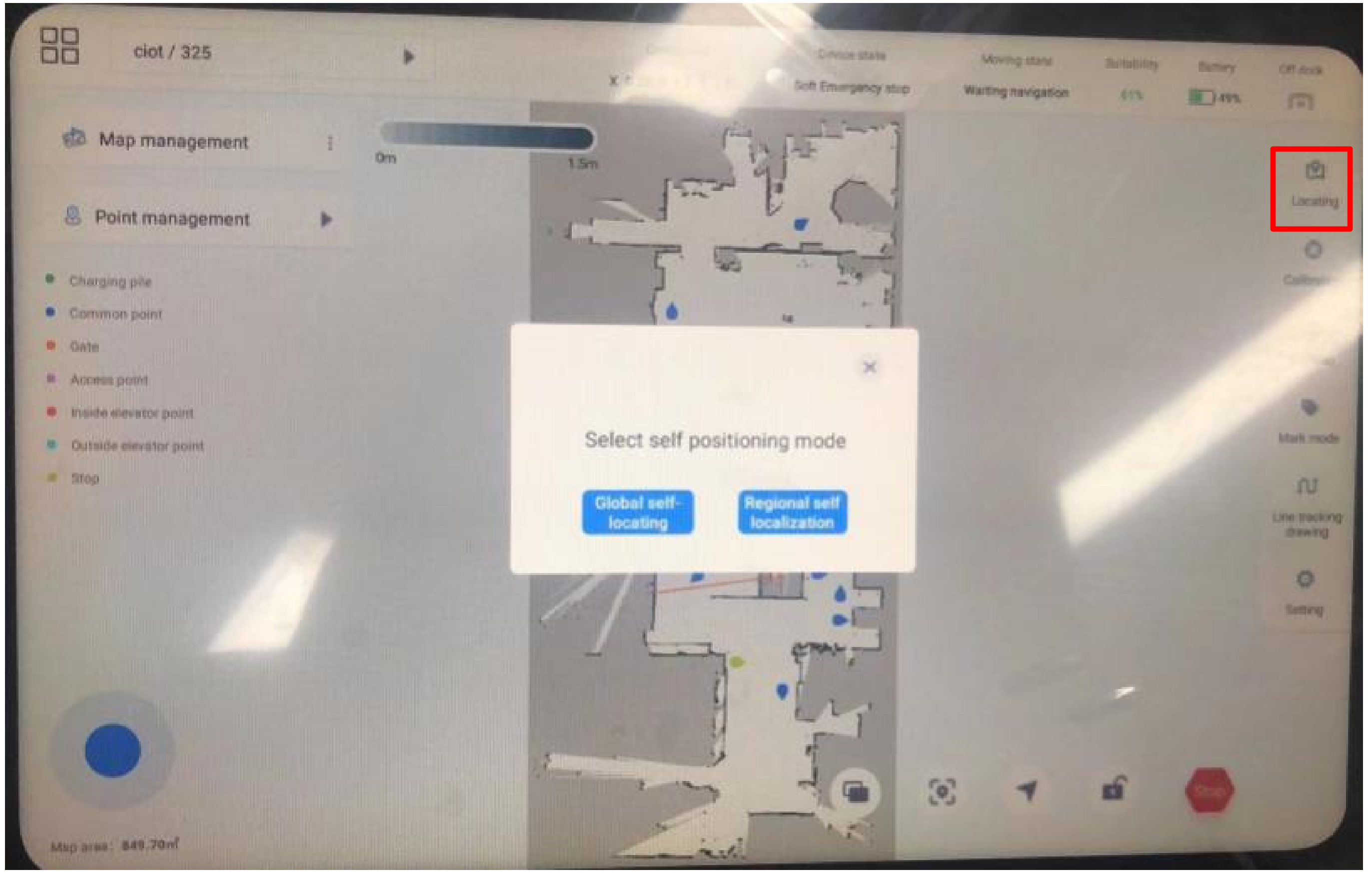
This section includes map layer, self-positioning, calibration, map editing, calibration mode and Settings.

General purpose (self - developed) chassis drawing tool

■ Global self-positioning module

Click the "Self-positioning" button, and a pop-up window appears as shown in the picture. Select "global self-positioning" and "regional self-positioning".

Click the "Global Self-positioning" button to start the global self-positioning, and the robot's position in the map can be automatically calibrated; After successful global positioning, a message is displayed indicating that the global self-location is successful.



Function description:

1. When the function is activated, personnel should stand behind the robot to operate. If the robot radar scanning area is blocked by standing in front, it is easy to cause positioning error.

2. When this function is started, the robot scanning environment should be selected as open as possible, with obvious features.

3. This function has a certain success rate and is time-consuming. It's not going to get it right every time. It was related to the environment and the size of the map.

4. If the positioning error occurs with this function, manual positioning power can be used to set the robot position

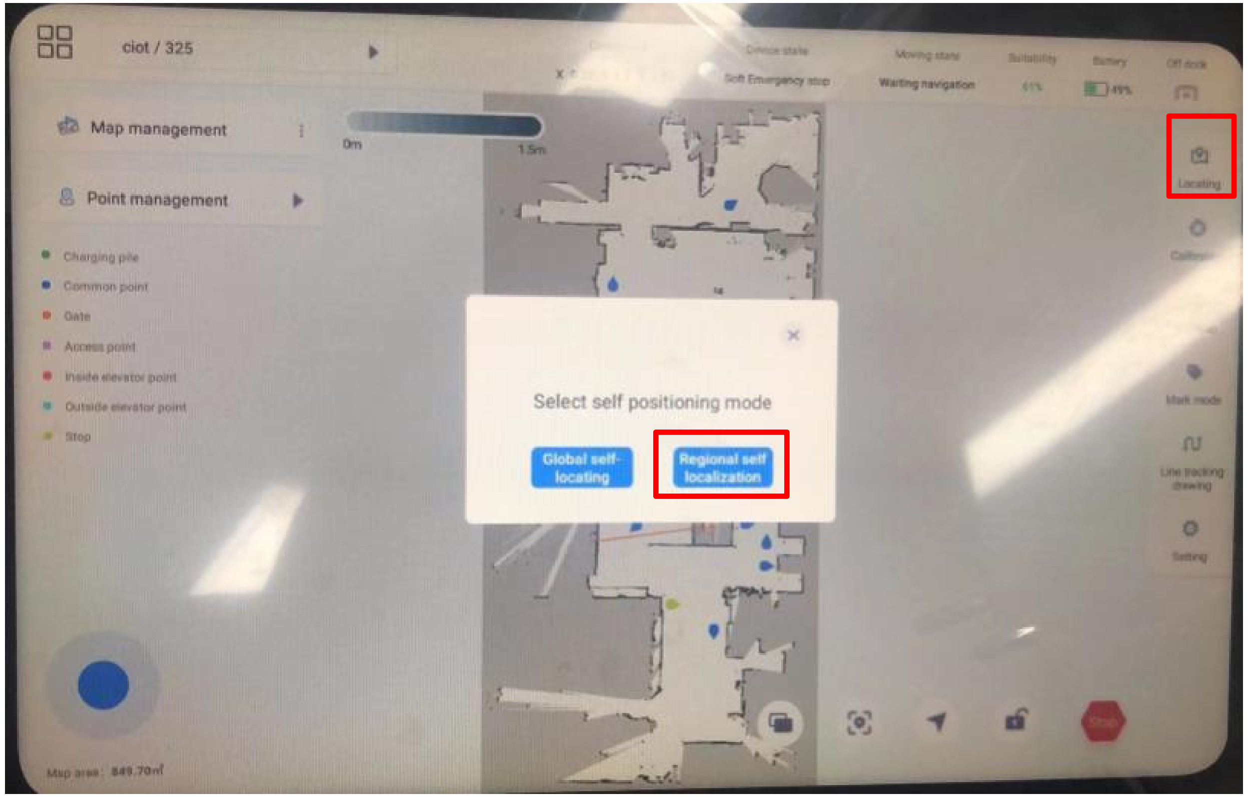
General purpose (self - developed) chassis drawing tool

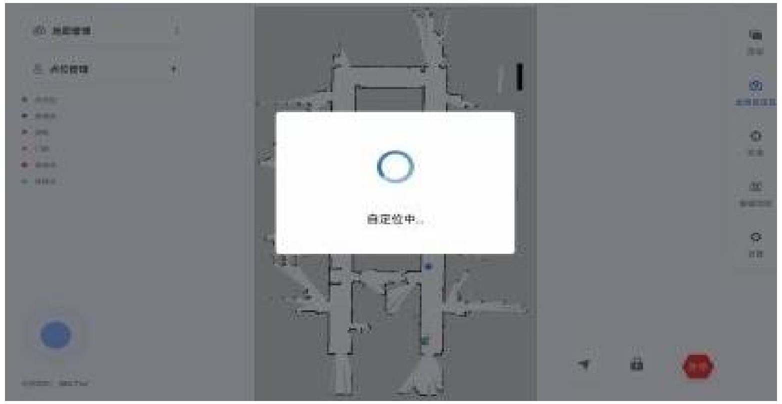
■ Global self-positioning module

Click the "Self-positioning" button, and a pop-up window appears as shown in the picture. Select "global self-positioning" and "regional self-positioning".

Click "region self-positioning" to pop up the page of selecting the region. Drag the border of the blue rectangle to zoom in and out and finish drawing the region

After the domain, click [OK] to start the region self-positioning. After the region self-positioning is successful, the page will prompt that the region self-positioning is successful





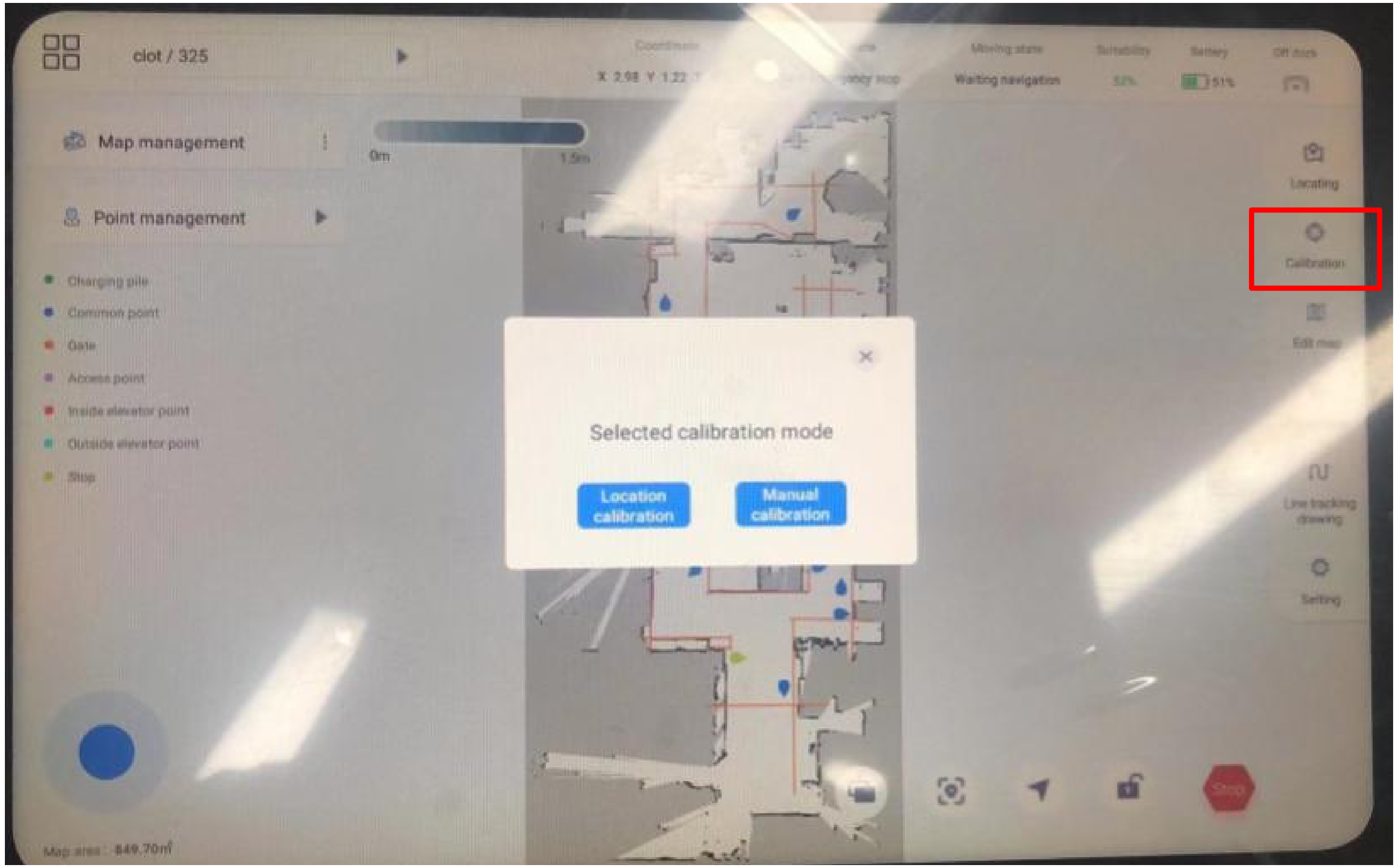
- **Function description：**

Select the actual position of the machine on the map and open the positioning, so that the positioning is more accurate and fast

General purpose (self - developed) chassis drawing tool

- Calibration module

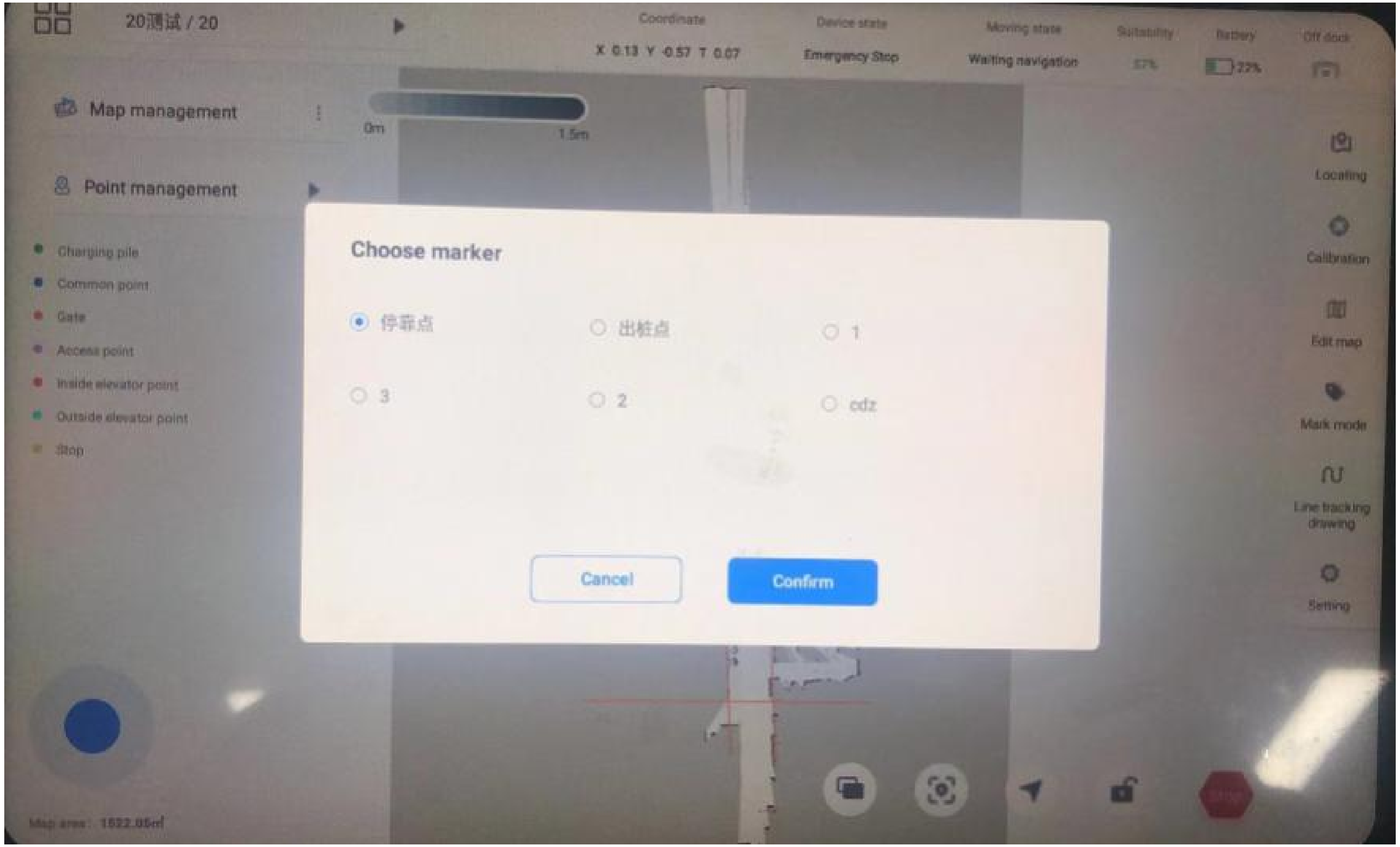
(1) Click the "Calibration" button, and a pop-up window appears as shown in the picture. There are two calibration methods: location calibration and manual calibration.



General purpose (self - developed) chassis drawing tool

- Calibration module

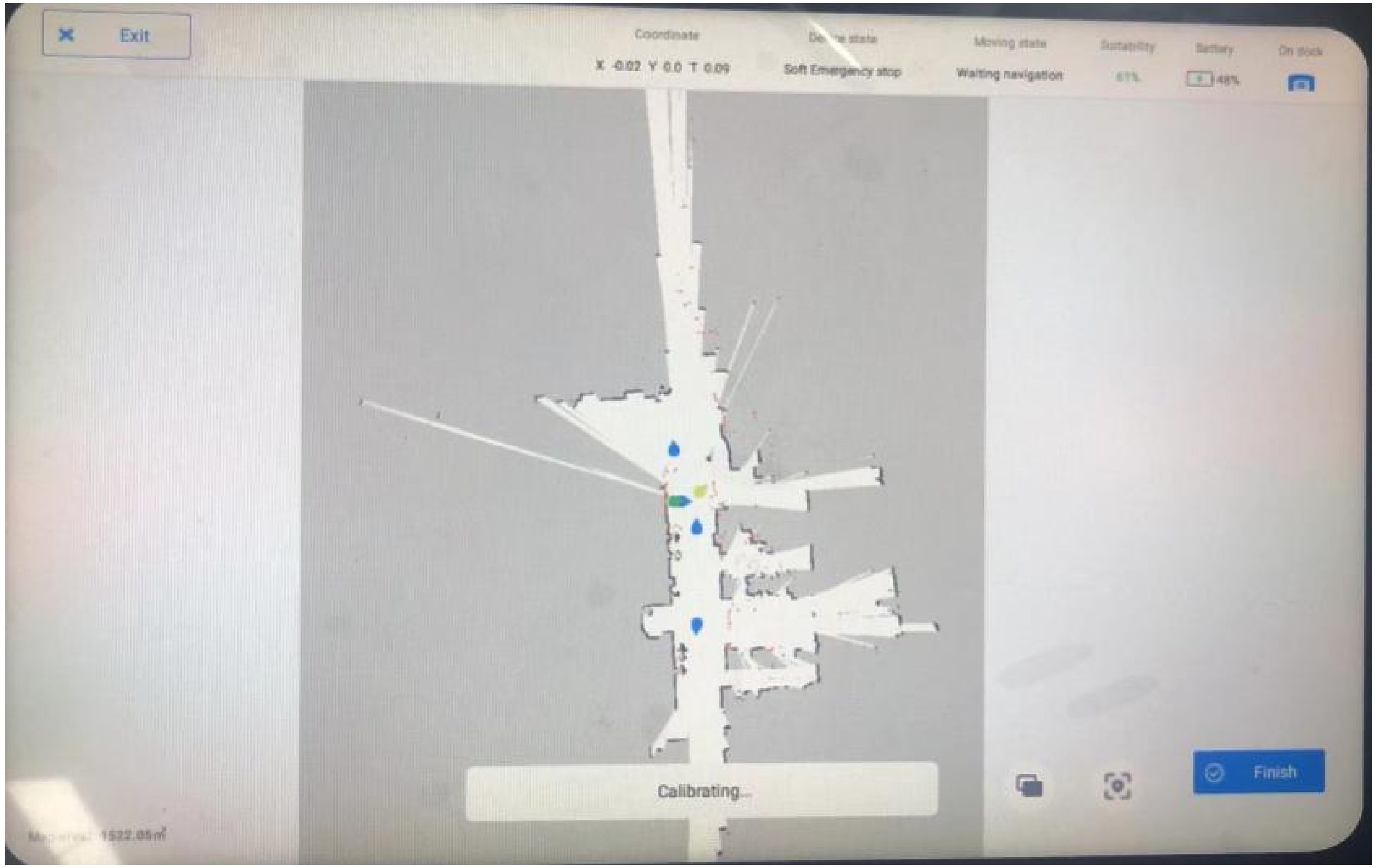
(2) Select "Location Calibration" to jump to the point list, select the point and click "OK" to start location calibration.



General purpose (self - developed) chassis drawing tool

- Calibration module

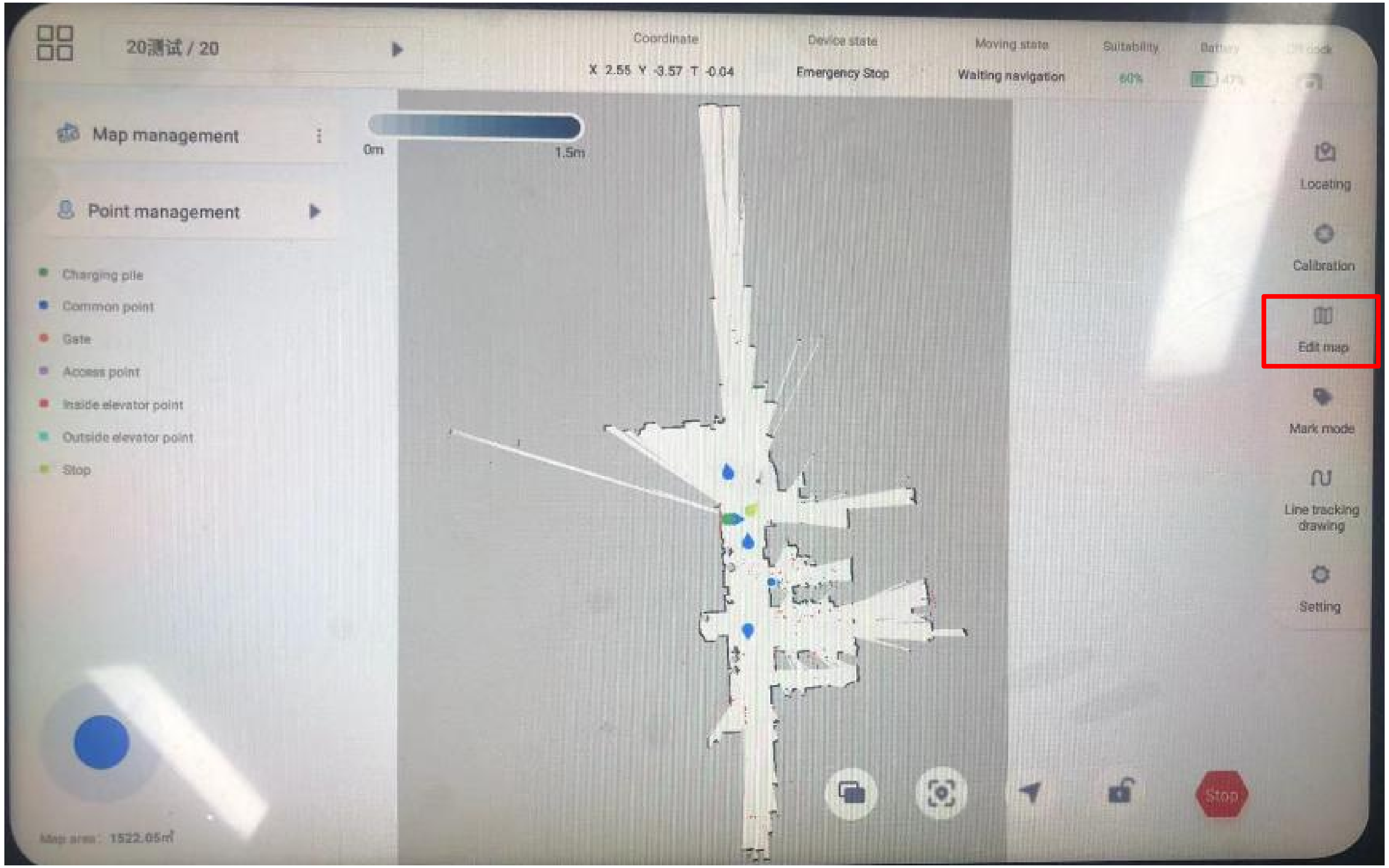
(3) Select "Manual calibration" to jump to the map page, select a point on the map, specify the direction, and start calibration.



General purpose (self - developed) chassis drawing tool

■ Edit map

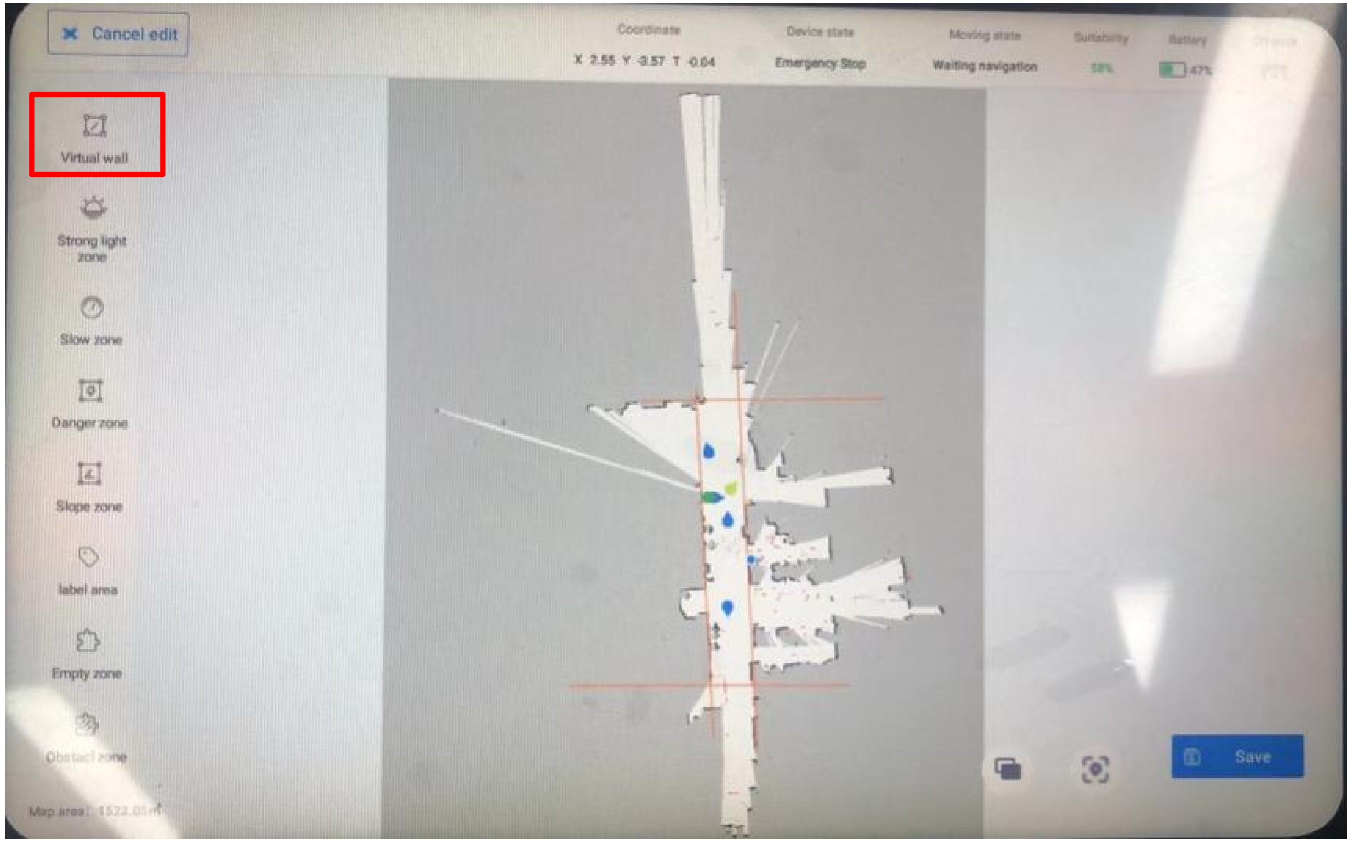
Click the "Edit Map" button to enter the edit map interface. Click Save after you finish drawing.

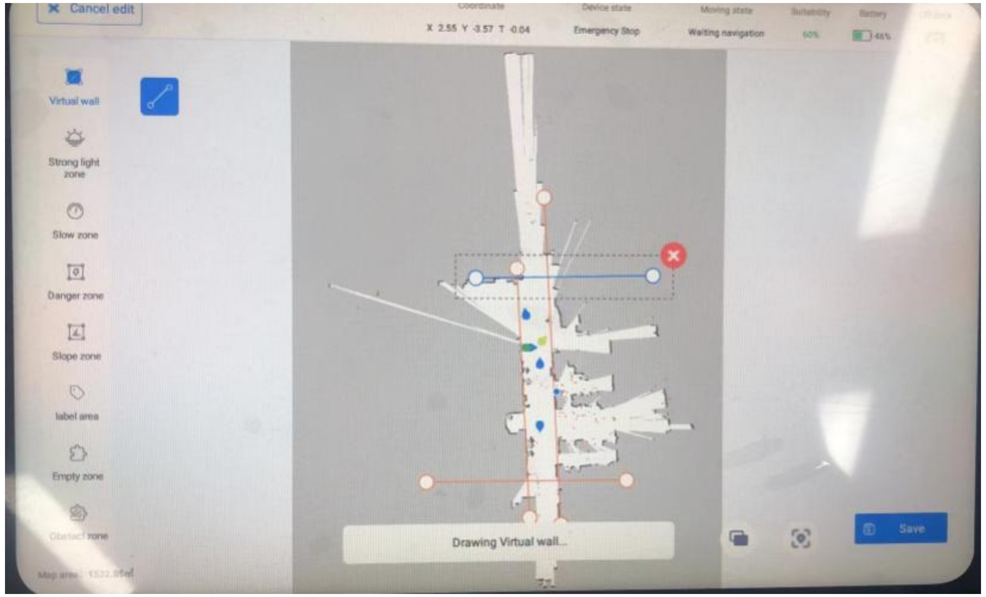


General purpose (self - developed) chassis drawing tool

■ Edit map

Virtual wall rendering：





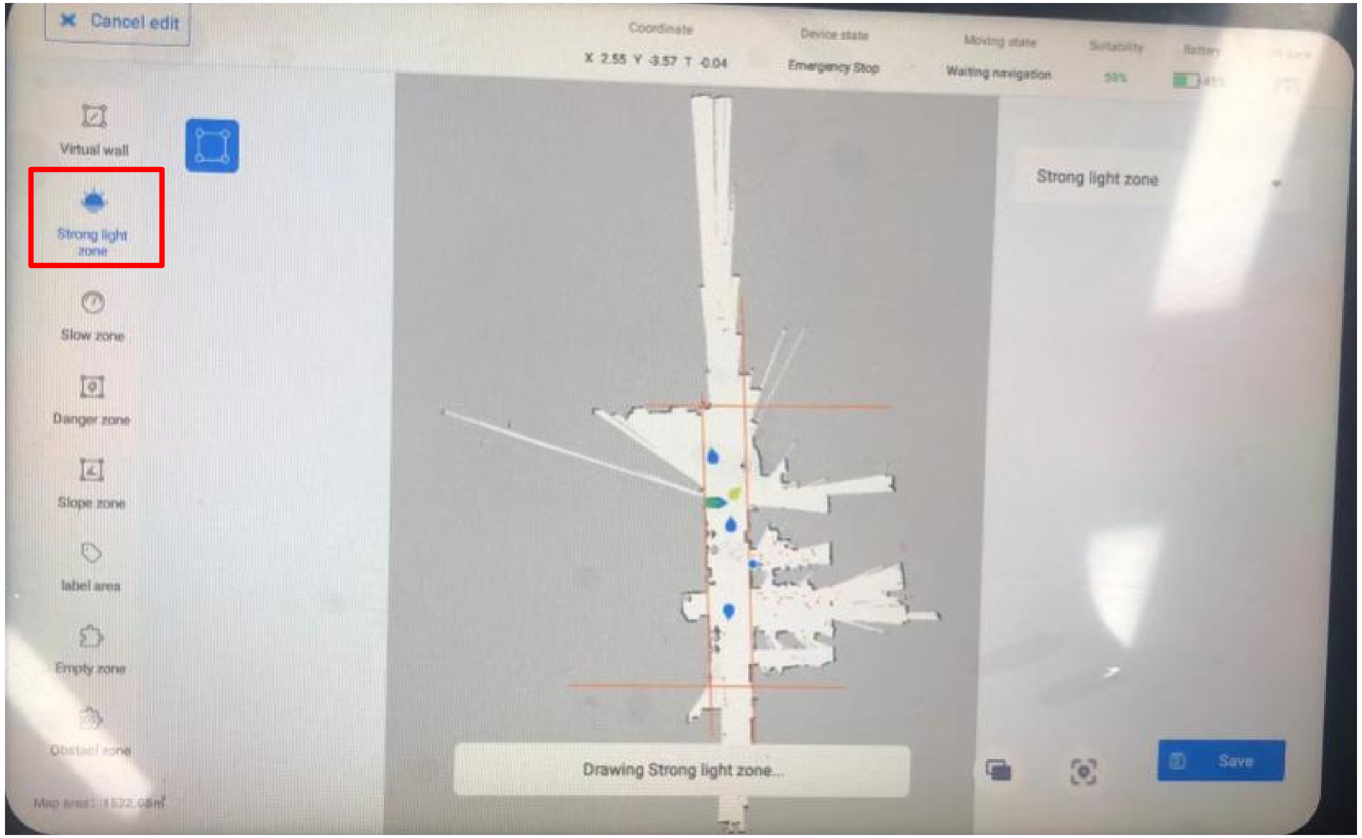
Click "Virtual Wall" to enter the virtual wall drawing page， select Start drawing the line segment. 

Select the virtual wall that has been drawn in the map, and an editable operation box will appear. Drag the dots at both ends of the line segment as shown in the figure to modify the length and position of the line segment. Click to delete the virtual wall. 

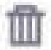
General purpose (self - developed) chassis drawing tool

- **Edit map**

Strong light interference area drawing:



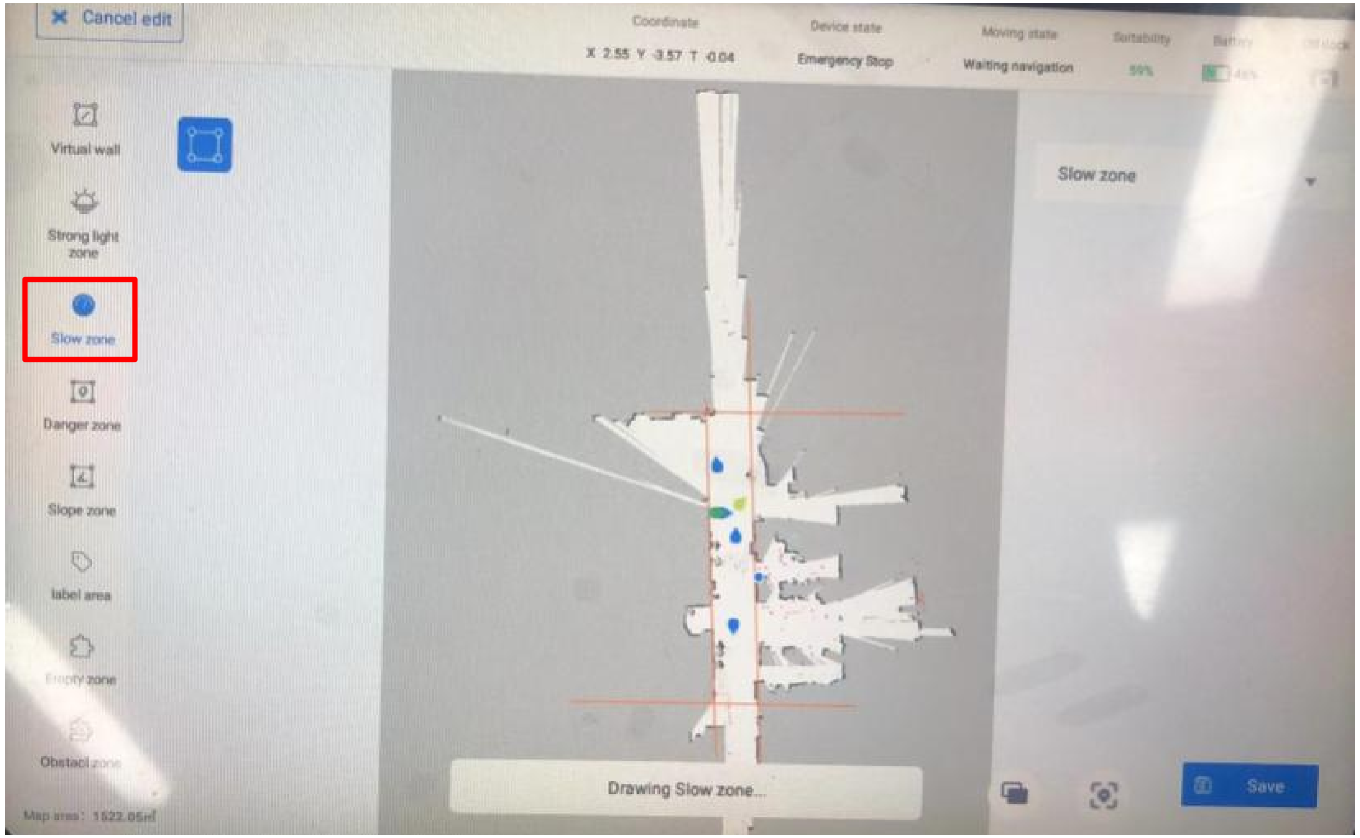
When selected, the corresponding workspace can be drawn (note: the workspace must be a closed area). When drawing the workspace, select the point position in a clockwise direction. 

Select the area in the map to remove the strong light interference. 

General purpose (self - developed) chassis drawing tool

■ Edit map

Deceleration zone rendering:



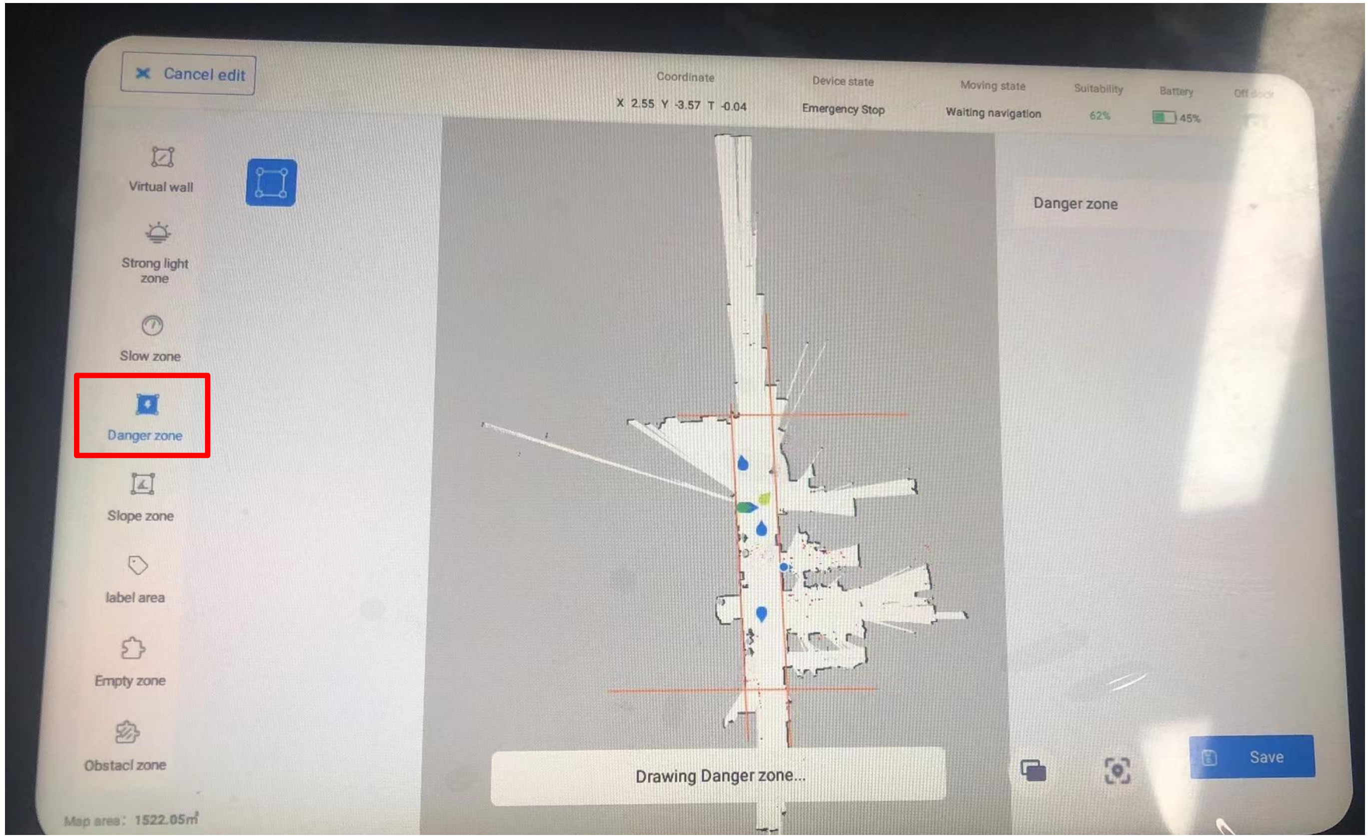
When selected, the corresponding workspace can be drawn (note: the workspace must be a closed area). When drawing the workspace, select the point position in a clockwise direction 

Check in the map to remove the slowdown zone 

General purpose (self - developed) chassis drawing tool

■ Edit map

Hazard mapping:



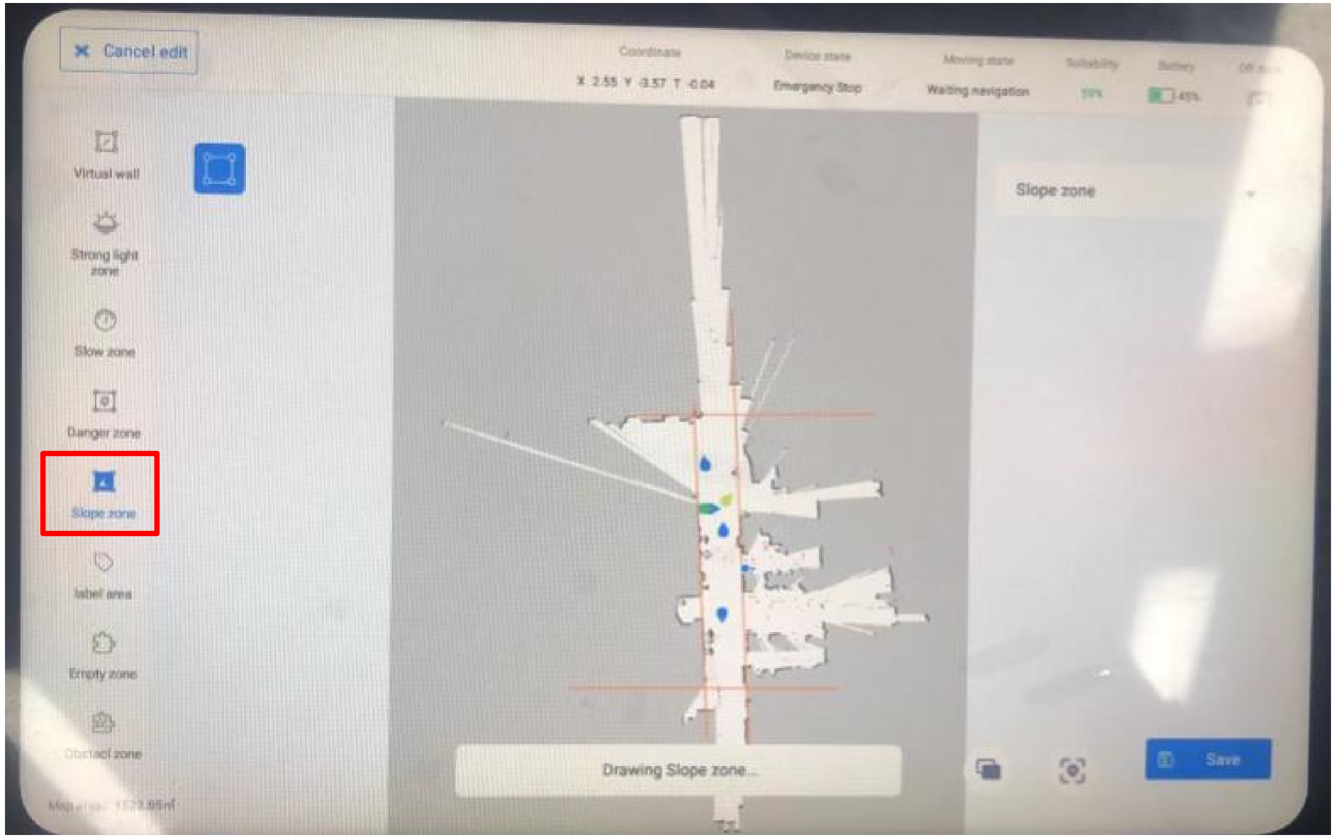
When selected, the corresponding workspace can be drawn (note: the workspace must be a closed area). When drawing the workspace, select the point position in a clockwise direction

Click on the map to delete the danger zone.

General purpose (self - developed) chassis drawing tool

■ Edit map

Broken zone drawing：



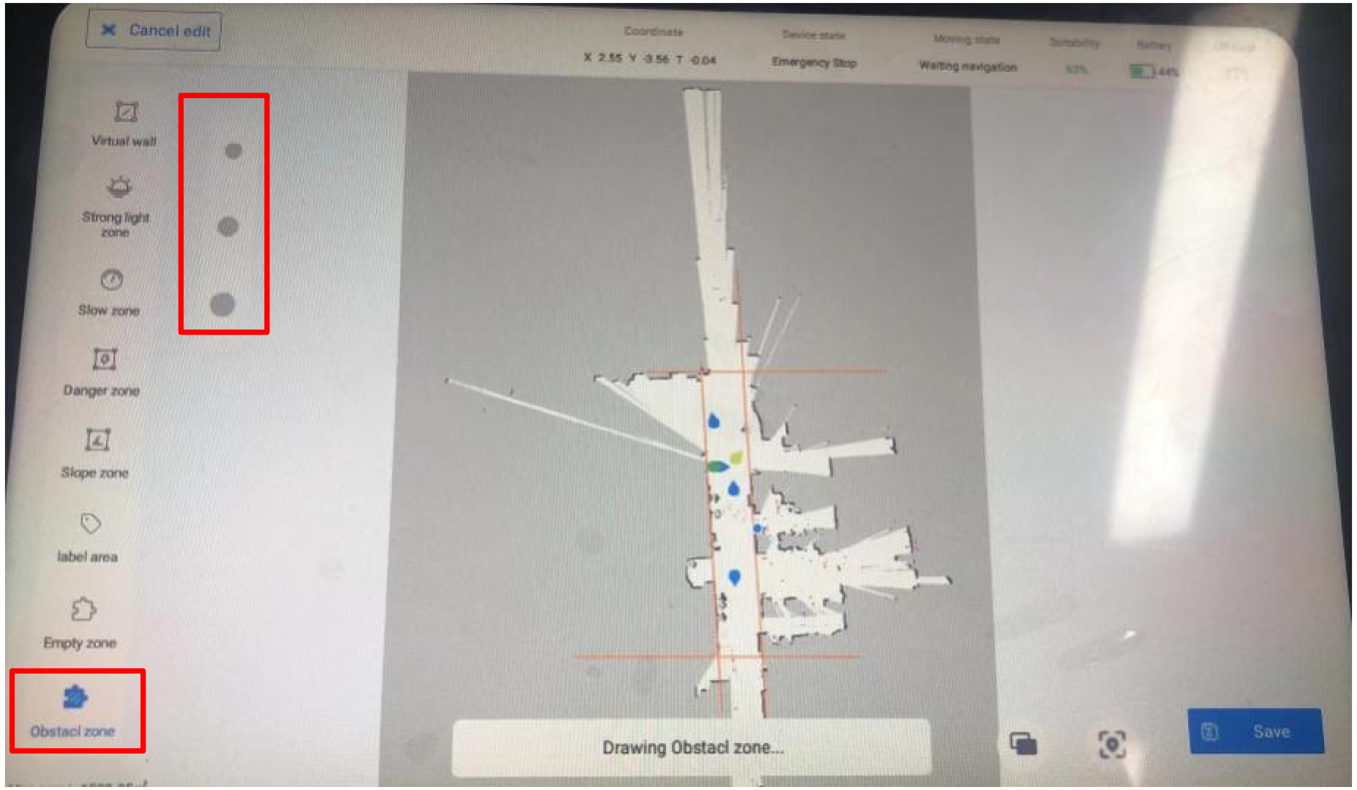
When selected, the corresponding workspace can be drawn (note: the workspace must be a closed area). When drawing the workspace, select the point position in a clockwise direction

Select the ramp area in the map

General purpose (self - developed) chassis drawing tool

■ Edit map

Obstacl zone:

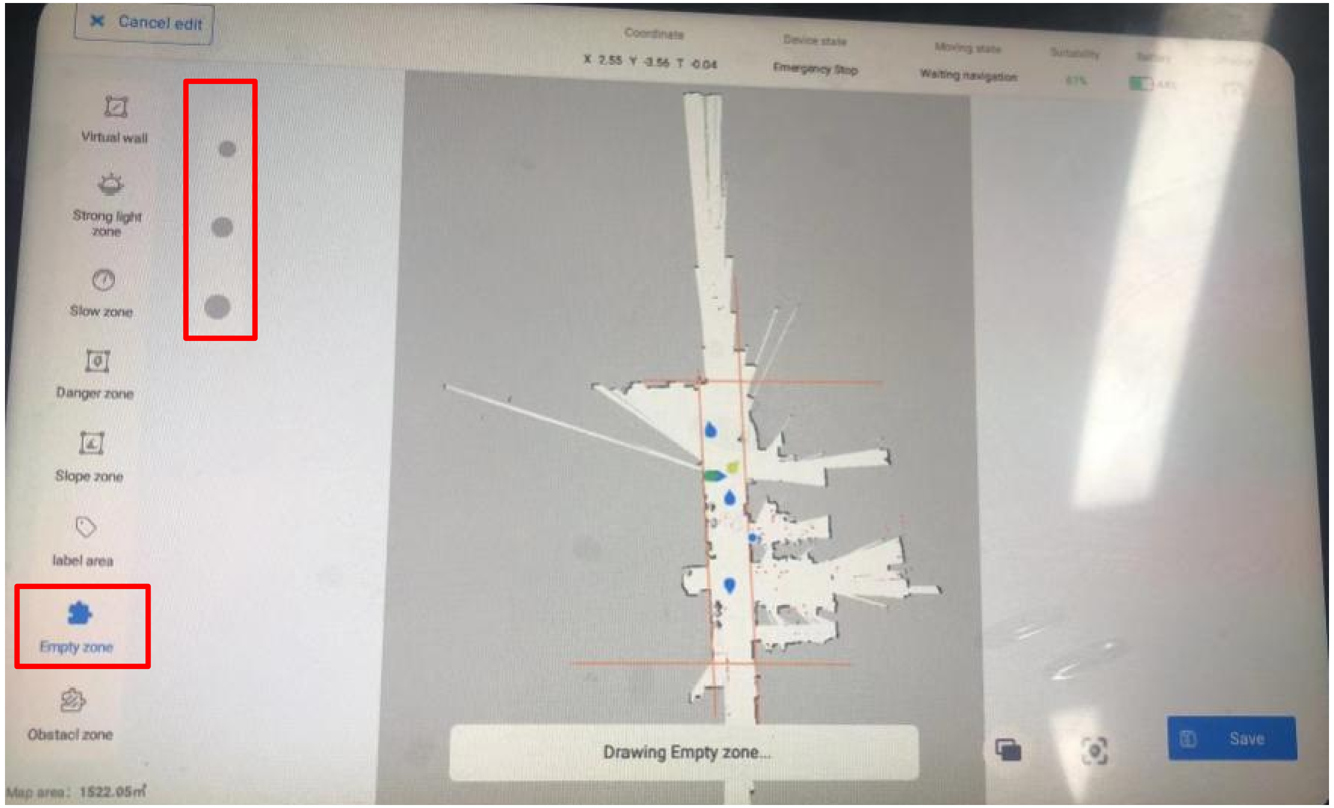


Click the obstacle area to enter the area to draw the page. After selecting the size of the brush, you can paint the appropriate area.

General purpose (self - developed) chassis drawing tool

■ Edit map

Empty zone:

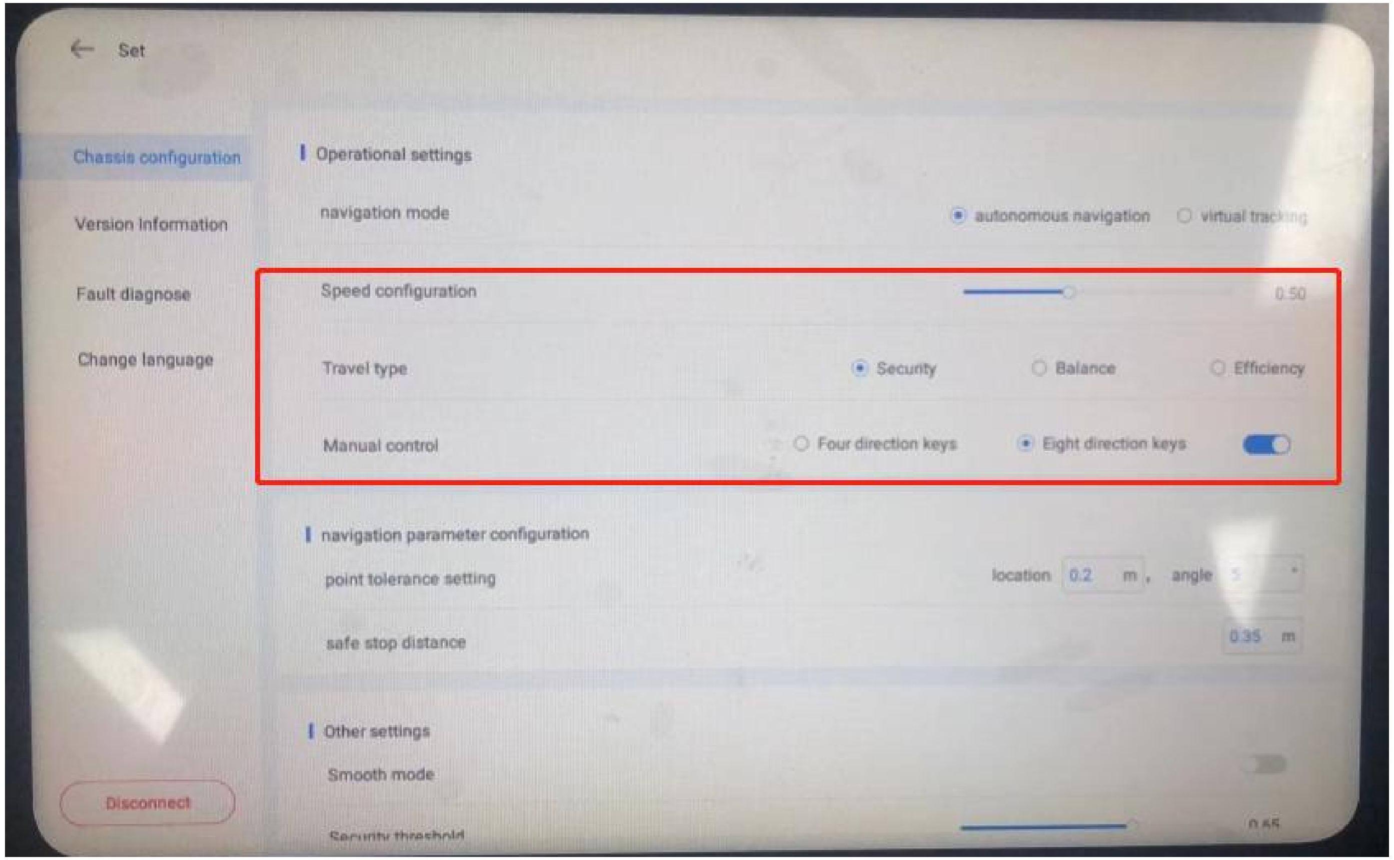


Click the blank area to enter the area to draw the page. After selecting the size of the brush, you can paint the appropriate area.

General purpose (self - developed) chassis drawing tool

■ Setting module

The setting module includes chassis configuration, version information, fault diagnosis, and language switching.



Chassis configuration:

(1) Operation setting:

Operation Settings include speed configuration, traveling mode and manual remote control. In the speed setting, the speed gear will only affect the speed of autonomous navigation of the robot but not the manual operation

Speed gear. - High: Fast motion(0.8m/s)

Speed gear. - Medium: Moderate movement speed (default robot speed 0.5m/s)

Speed gear - Low: The movement speed is slow (0.3m/s)

Travel mode - Safety: obstacle avoidance distance is large, not too narrow laneTravel Mode - Balance: Obstacle avoidance distance is moderate

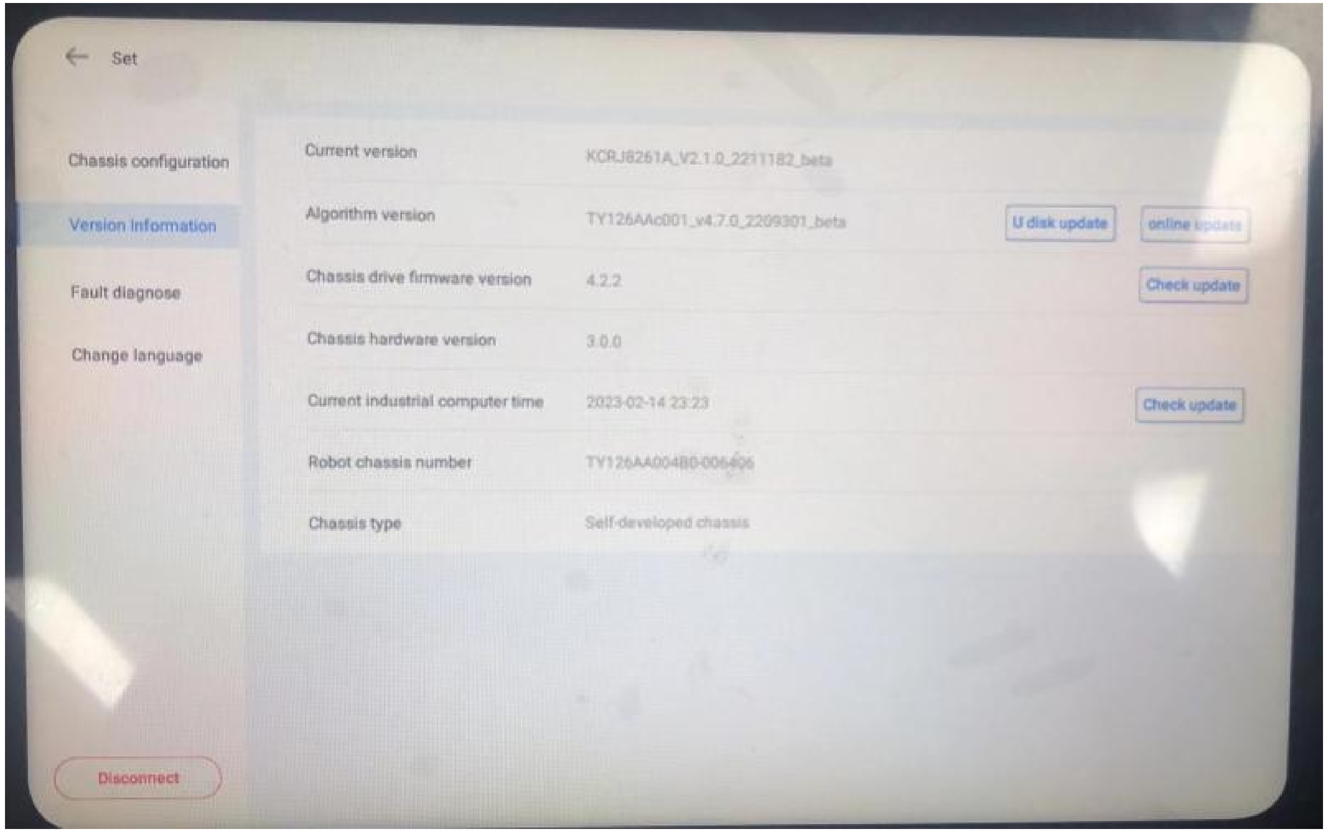
Travel mode - Efficiency: Obstacle avoidance distance is small, can be too narrow lane

Directional Mode - Four arrow keys: Traditional arrow keys figure: 

Directional Mode - Eight Arrow Keys: Traditional joystick, figure: 

General purpose (self - developed) chassis drawing tool

■ Version information

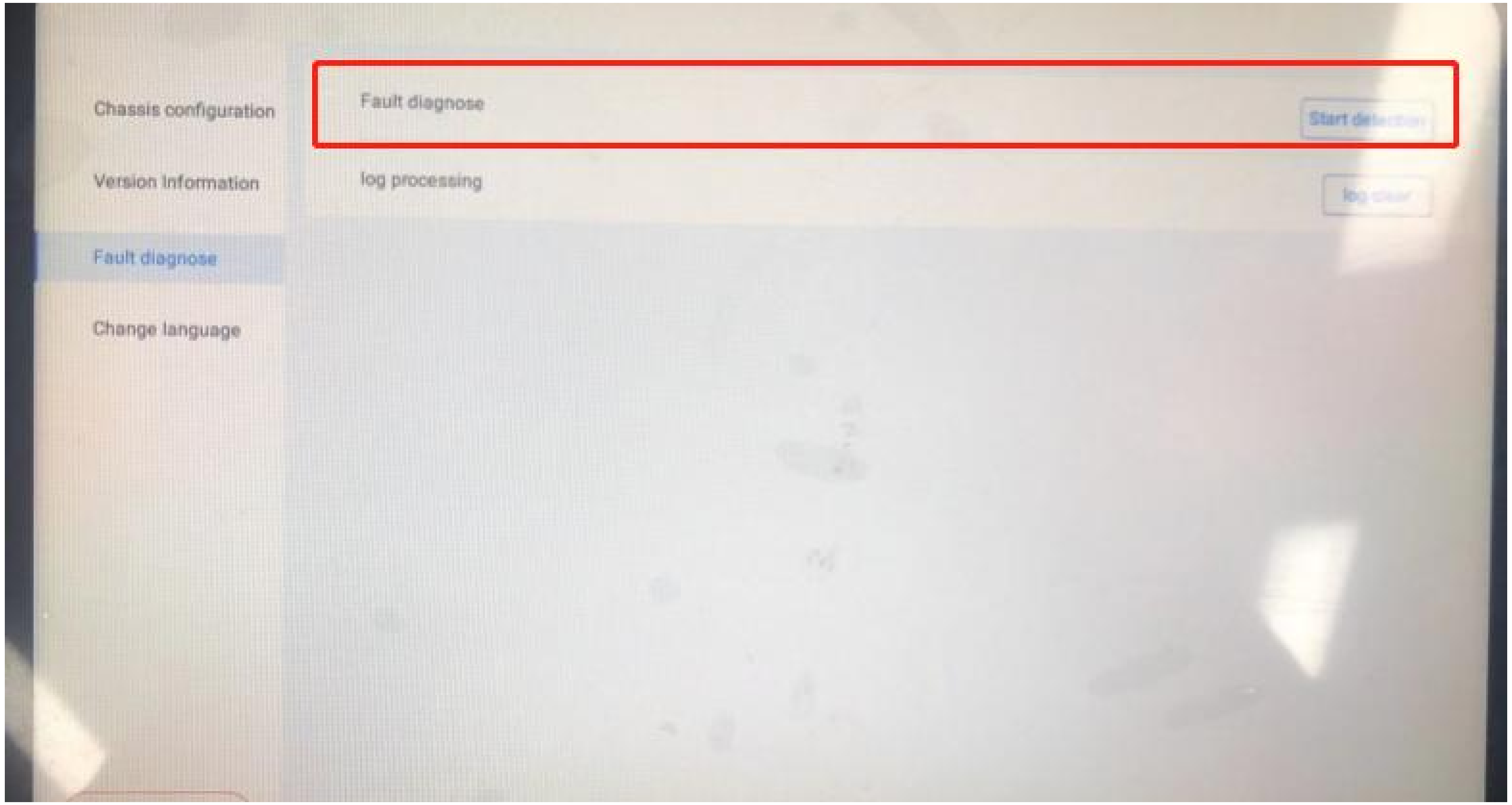


- Support algorithm version [U disk], [online] two update methods, support the bottom disk drive firmware update, industrial computer time update.
- After the chassis algorithm is updated, please shut down and restart the robot.
- After the underlying firmware version is updated, the robot will automatically power off. Do not do anything before the robot automatically power off and power off. After the robot automatically shuts down, the robot can be turned on normally.
- After updating the industrial computer time, please shut down and restart the robot.

General purpose (self - developed) chassis drawing tool

■ Fault diagnosis

Click [Start detection] to start the self-detection and enter the page of self-detection results to view the detection results.

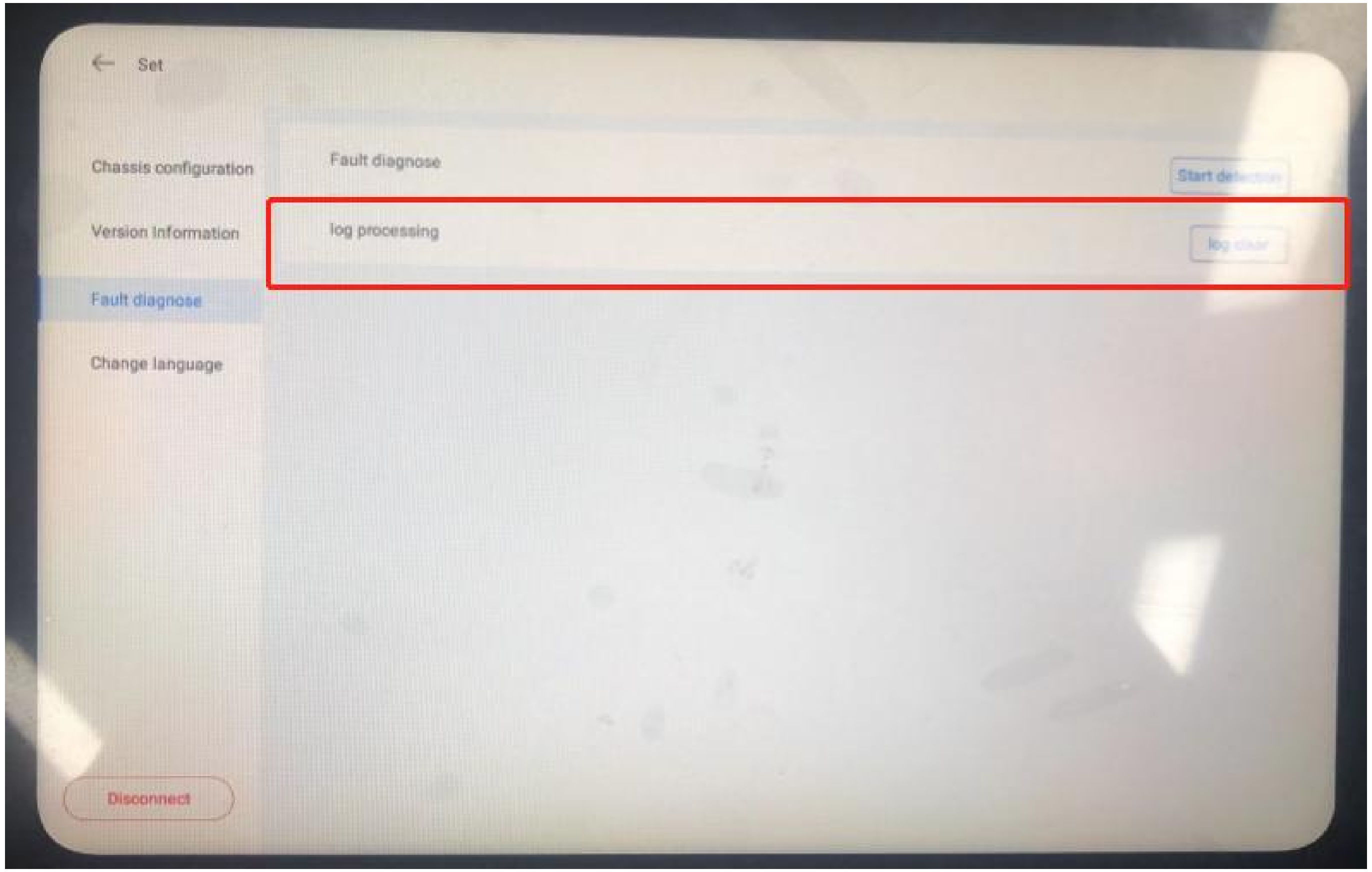


General purpose (self - developed) chassis drawing tool

■ Fault diagnosis

Click "Log Clear" to clear the log files in the industrial computer and free the memory space of the industrial computer.

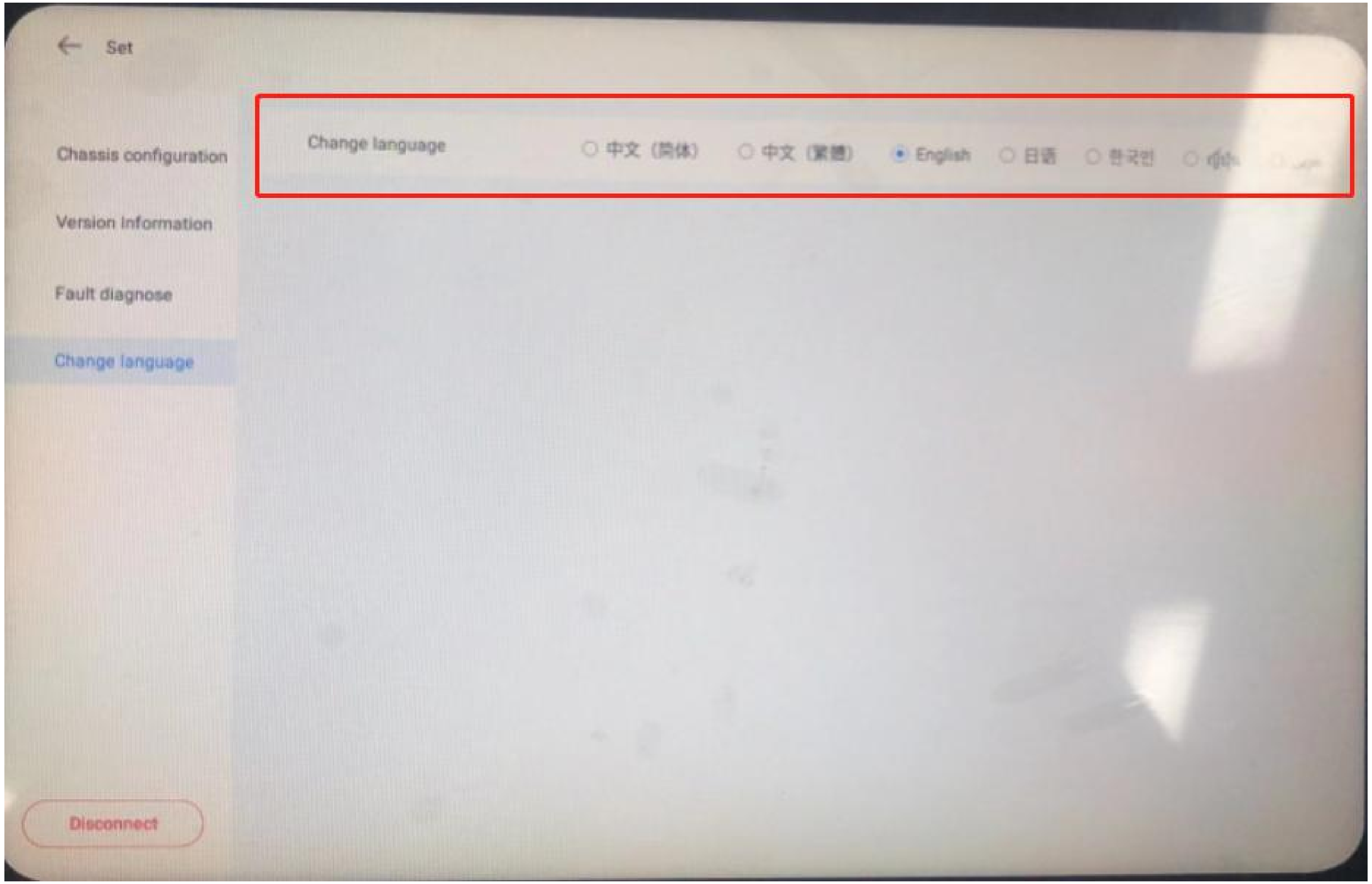
If the log memory usage is excessive, the chassis startup is affected or the chassis is stuck, you can clear logs.



General purpose (self - developed) chassis drawing tool

■ Language switching

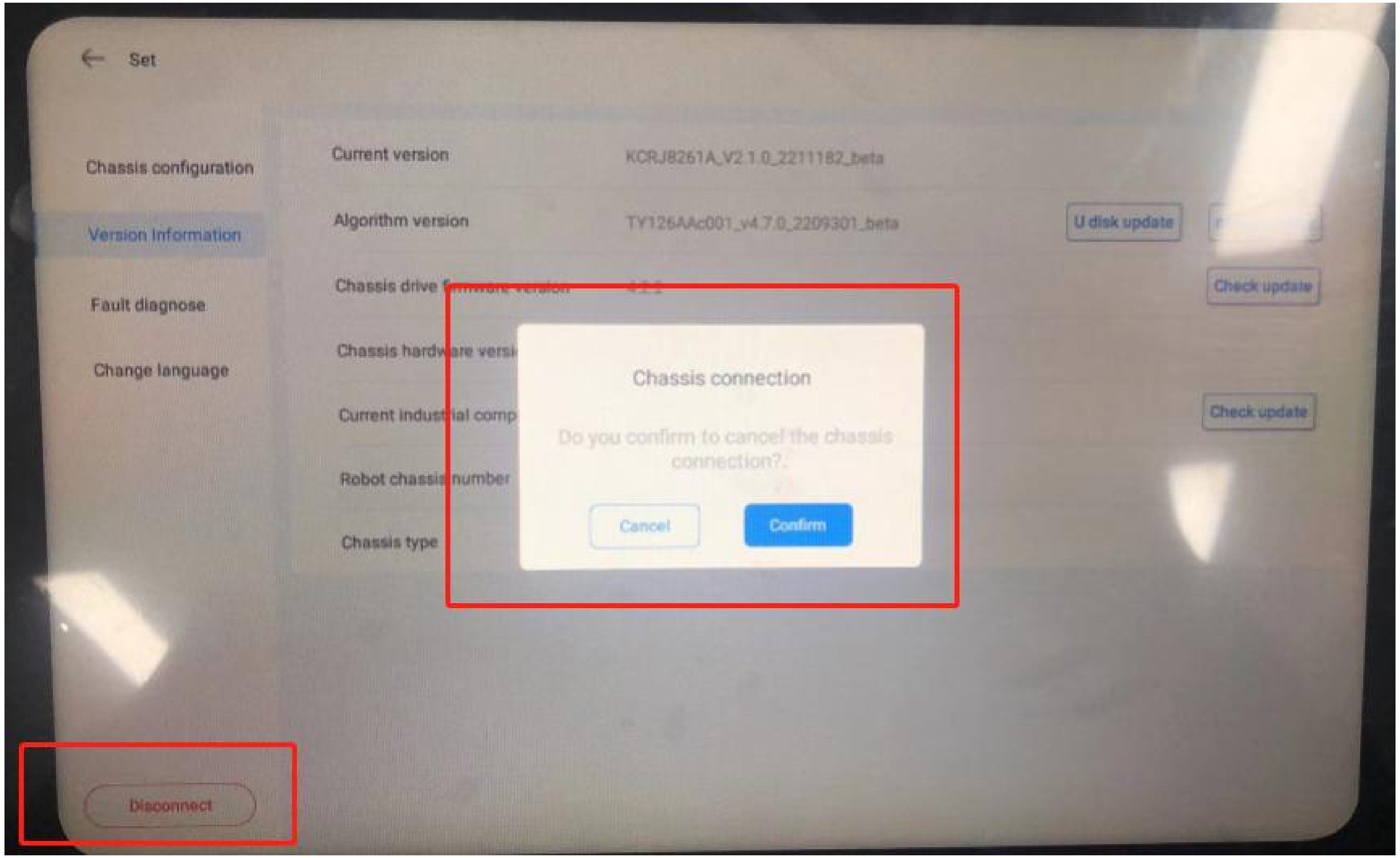
If you click a different language, different languages will be displayed in the deployment software.



General purpose (self - developed) chassis drawing tool

■ Disconnect

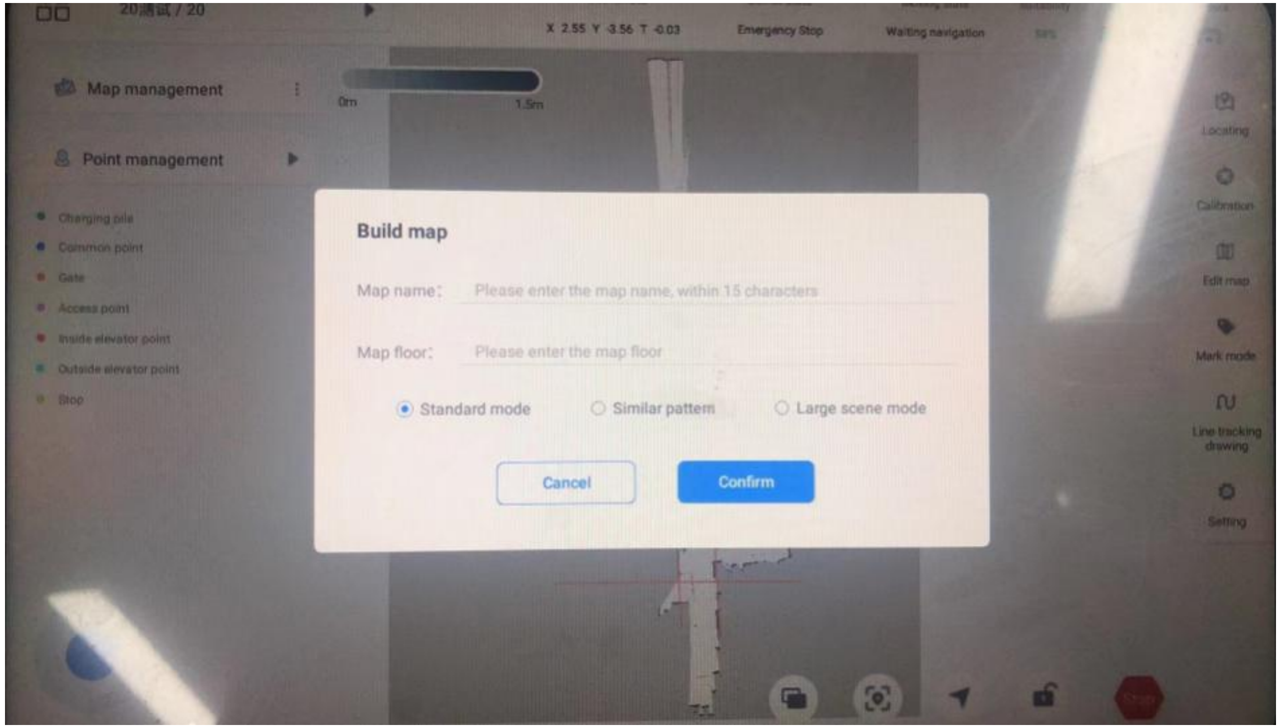
Click Disconnect to disconnect from the chassis.



General purpose (self - developed) chassis drawing tool

■ Start illustration (self-developed and deployed APP)

Click the button and click New Map to enter the New Map page. Fill in [Map Name], [Map Floor], and select [Map Mode].



Standard scenario:

Standard scenarios are used in drawing construction

Similar scene:

If the drawing environment is mostly similar corridors, narrow lanes and other scenes with small changes in reference objects, it is recommended to use the "similar scene" mode for drawing construction。

Big scene:

If the drawing environment is larger than 1000 square meters or is relatively empty, you are advised to use the Large scene mode to build the drawing.